

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE CONTINUOUS ASSESSMENT TEST - II - JUNE-2026 CET 05201 - COMPUTER NETWORKS

SEMESTER: II

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

PART - A (Answer all the Questions) (7 x 2 = 14)

1. Give two differences between UDP and TCP.
2. What is different between Ethernet Cable and stack cable?
3. Mention two functions of presentation layer.
4. Mention two routing protocol used in Network Layer.
5. Differentiate cross over and straight through cable.
6. What is Router and Routing table?
7. Why do network switches use MAC addresses for forwarding traffic while routers use IP addresses for routing?

PART - B (Answer any three Questions) (3 x 4 = 12)

8. Describe major four (4) functions performed by a router in a computer network.
9. Elaborate the Pros of OSI Model.
10. Compare and contrast a Switch and a Router in terms of operation, addressing, OSI layer, and applications.
11. Identify the two sublayers of the Data Link Layer and explain the role of each sublayer in network communication.
12. A network administrator needs to send data to a single device, a group of devices, and all devices on a network. Explain the types of MAC Addresses involved in each case with examples.
13. What is a Routing Protocol? Write short notes on OSPF, BGP, and EIGRP.

PART - C (Answer any two Questions) (2 x 12 = 24)

14. With the aid of examples, analyze the internal architecture of a router and discuss the functions and applications of different router types.
15. An entrepreneur is setting up an internet café and needs a device that can intelligently forward frames to the correct computers without causing unnecessary network traffic.
 - i. Define the network device described above.
 - ii. Discuss the different types of this device and their applications.
16. Identify and explain the tool used for crimping network cables. Furthermore, describe the importance and functions of network cable testing and troubleshooting tools.
17. Explain the seven layers of the OSI model with the help of diagram. For each layer;
 - a) State its name
 - b) Describe its function in network communication
 - c) Give one example of a device, protocol, or technology related to that layer.